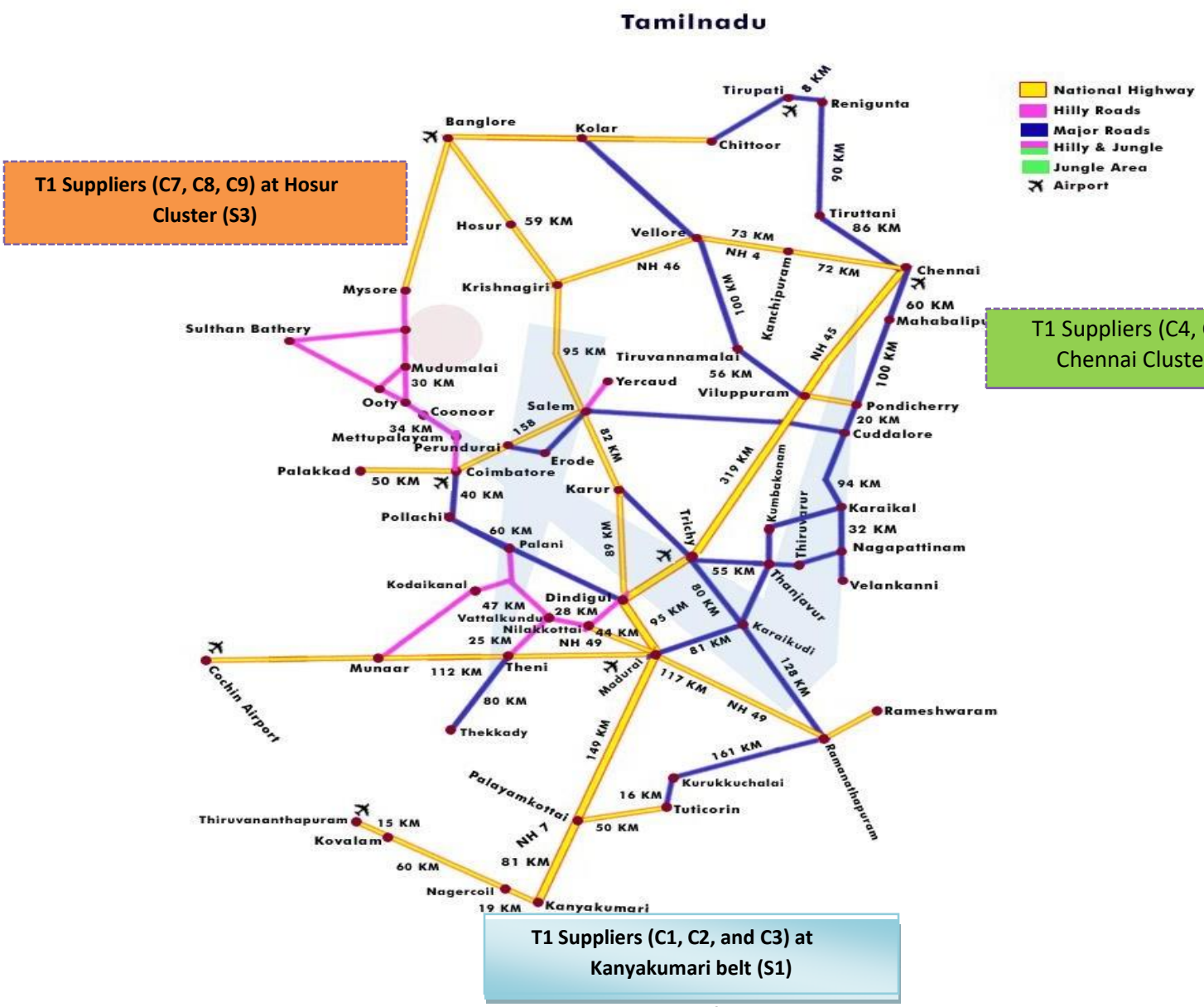
SUPPLY CHAIN NETWORK DESIGN



Do not use distance data given in the map. Please refer Google Maps to determine actual distance between any two nodal points.

Average Speed in NH = 250 KM /day; Hilly Roads =150 KM/day; Major Roads = 200 KM/day; Jungle Area = 100 KM / Day; Hilly & Jungle = 75 KM/Day

Your new Farm Equipment company will adopt *Assemble-to-Order* (ATO), which means your firm doesn't stock FG inventory. On receipt of orders from Dealers the company starts assembly and shipment. Hence, material shall be stored at different locations across Tamilnadu. Dealer locations and demand potential – Karaikudi (240 units pm), Coimbatore (300 units pm), Tirupati (160 units pm), Rameshwaram (140 units pm), Tuticorin (120 units pm), Cochin (220 units pm), Pondicherry (140 units pm) and Thiruvanthapuram (100 units per month), for next 4-5 years. INR The company emphasizes both efficiency and responsiveness equally.

Cost of material transportation (Suppliers-to-Stores) per SKU[1] per KM = Rs. 0.5;
Cost of material transportation (Stores-to-Assembly Plant) per SKU per KM = Rs. 0.5;
Cost of FG transportation (Plant –to-Dealer) per shipment per KM = INR 100; truck capacity = 3 FG unitsService Level to be maintained or Order-to-Delivery(OTD): Assembly Plant to Dealer = 60 Hours Max.
Adopts *Just-in-Time* Production; Order wait Time for Production: 4-6 Hours; Assembly Takt Time = 2 Hours per Unit;

Special Economic Zones: Madurai, Trichy, Dindigul; Locating warehouse in these SEZs will reduce space cost by half.

Processing Costs at Assembly Units:

Up to 100 units = INR 300 per unit; 101 to 200 units= INR 200 per unit; 201 to 400 units= INR 100 per unitMax. Plant capacity can be up to 400 Units per month; Min. plant capacity should be 40 units per month Minimum cost per Assembly Unit per month = INR 12000 {for 0 to 40 SKUs}

Inventory Policy and cost at Stores:

1 SKU = 3 C9 = 4 C8 = 5 C7 = 6 C6 = 7 C5 = 8 C4 = 9 C3 = 10 C2 = 11 C1

Size of one Material Pallet SKU = 5 Sq. Ft; The stores have max. 10 –level storage racks
Storage Costs (space cost): INR 1 per Sq. Ft per day; i.e., for 5 square foot = INR 5 per Sq.Ft per day (10 SKUs can be stored in 5 Square foot);
In SEZs, space cost is INR 0.5 per Sq. Ft per day; i.e., for 5 square feet = INR 2.5 per Sq. Ft per day

Constraints:

As a policy, material is replenished at stores once in every 6 days
Assembly plants and component stores should be located at any nodal point (identified place) in above map.
Management has sanctioned setting up maximum 5 assembly units and 5 material stores.

For Example:

If a store is serving Assembly Unit 1 (200 units per month) and Assembly Unit 2 (120 per month),Then the store has to carry stock of 320 units of each *Component* per month.
AVG aggregate demand per day = 320/30 = 11 units (each component)
Stock carried (of each component) = AVG aggregate demand per day x 6 days = 11x 6 = 66 units

	AVG Consumption (during replenishment period)	No. of SKUs
C1	66	6
C2	66	7
C3	66	7
C4	66	8
C5	66	9
C6	66	11
C7	66	13
C8	66	17
C9	66	22
Total No.	594	100

Stock carried (All components at each store) =66 x 9 components = 594 units which requires 100 SKUsHence the capacity of the store should be at-least = (100 SKUs /10) *25 = 250 Square Feet space
Since space is required for aisle, storage of equipment, inward and outward dock areas
Approx. in farm equipment industry 3 times storage space is required to accommodate, so the total space requirement for this store = 3 x 250 = 750 Square foot, which means
As per square foot cost is INR 1 per day; for 750 Sq. ft. per day, cost is INR 750 per dayPer month cost = 750 x 30 = INR 22500 per month; in SEZs = INR 11250 per month

Transportation Strategy:

1. LTL[2] (less than truck load) is used for shipment between upstream stages in the supply chain for efficiency;
2. FTL[3] (full truck load) is used from AU to Dealers to ensure responsiveness.

[1] SKU– Stock Keeping Unit –Specific no. of same product units packaged into one unit that can be handled at a time.
[2] Less than truck load (carriers which charge shipping tariff @ per unit basis)
[3] Full truckload (FTL) carriers normally deliver a semi-trailer to a shipper who will fill the trailer with freight for one destination.

DISTANCE MATRIX								
	S1	S2	S3	AU1	AU2	AU3	AU4	AU5
W1								
W2								
W3								
W4								
W5								
Tirupati								
Coimbatore								
Karaikudi								
Thiruvananthapuram								
Rameshwaram								
Tuticorin								
Cochin								
Pondicherry								
		Total Distance Travelling:						

Maximum service time available from any AU to any Dealer is 2.5 days.
Since, Order Processing Time = 10 -12 hours; available time to ship from any AU to any Dealer is maximum 2 days.

In the following example, if the shortest route identified between an Assembly Unit and a Dealer is as shown below, then the maximum distance allowed within 2 days is 288 KMs through different types of roads.

Route	Distance (in KM)	Time Required (in Days)
National Highway (NH)	72	0.288
Major Roads(MR)	18	0.09
Hilly Roads (HR)	144	0.96
Jungle Area (JA)	24	0.24
Hilly and Jungle (H&J)	30	0.4
Total	288	1.978

USE DISTANCE & SPEED CALCULATOR

Allocation MATRIX								
	S1	**S2**	**S3**	**AU1**	**AU2**	**AU3**	**AU4**	**AU5**
W1								
W2								
W3								
W4								
W5								
Tirupati								
Coimbatore								
Karaikudi								
Thiruvananthapuram								
Rameshwaram								
Tuticorin								
Cochin								
Pondicherry								

SUPPLIER – TO – WAREHOUSE ALLOCATION CALCULATION:

Since components are replenished (from suppliers to warehouse) once in 6 days, then in a month there would be 5 replenishment cycles. If a warehouse(store) is serving two assembly plants (producing @ 100 units per month), then from any suppliers, 20 units of those components are replenished for every 6 days. For instance, from supplier sector (S3) to any warehouse, no. of SKUs shipped per cycle is 16 as shown below. Then no. of SKUs shipped per month = 16 SKUs x 5 replenishment cycles = 80 SKUs. **1 SKU = 3 C9 = 4 C8 = 5 C7 = 6 C6 = 7 C5 = 8 C4 = 9 C3 = 10 C2 = 11 C1**

Supplier (S3)	AVG Consumption (during Replenishment Period)	No. of SKUs
C7	20	4
C8	20	5
C9	20	7
Total No (per cycle)	60	16
Total (PM)	300	80

WAREHOUSE –TO-ASSEMBLY UNIT (AU) ALLOCATION CACULATION:

For example, if an Assembly Unit (AU) produces 40 units per month, then consumption of different components is as follows:

Sub-Assembly	AVG Consumption (during month)	No. of SKUs
C1	40	4
C2	40	4
C3	40	4
C4	40	5
C5	40	6
C6	40	7
C7	40	8
C8	40	10
C9	40	13
Total No.	360	61

No. of SKUs shipped from Warehouse(s) to the Assembly Unit (AU) is 61 SKUs. Again this quantity might be shared across more than one warehouse.

ASSEMBLY PLANT TO DEALER ALLOCATION CALCULATION: Number of units that an Assembly Unit serves dealer(s) per month is the actual capacity of the respective AU which is shared across the corresponding dealers.